Pneumonia Prediction Utility

By: Tyler Haisman

Overview

My pneumonia prediction utility utilizes deep learning models to analyze chest X-ray images and predict whether pneumonia is present and, if so, whether it is bacterial or viral in nature.

Technology Stack

Frontend:

- Framework: Next.js (React)
- Styling: Tailwind CSS
- Language: TypeScript

Backend:

- Framework: Django (Python)
- Deep Learning Library: TensorFlow Keras

Models

1. Chest X-ray Detection Model:

- This model determines whether the image is a chest X-ray or not.
- Trained on nearly 2000 chest X-ray images.
- Utilizes a convolutional neural network (CNN) architecture.

2. Pneumonia Classification Model:

- This model identifies if pneumonia is present and categorizes it as bacterial or viral.
- Trained on over 4000 chest X-ray images depicting normal, bacterial pneumonia, and viral pneumonia cases.
- Utilizes a CNN architecture.

Data Sources

Chest X-Ray Images (Pneumonia) Dataset:

- Kaggle Dataset: Chest X-Ray Images (Pneumonia)
- Contains chest X-ray images depicting normal, bacterial pneumonia, and viral pneumonia cases.
- Kermany, Daniel; Zhang, Kang; Goldbaum, Michael (2018), "Labeled Optical Coherence Tomography (OCT) and Chest X-Ray Images for Classification", Mendeley Data, V2, doi: 10.17632/rscbjbr9sj.2

Random Images Dataset:

- Kaggle Dataset: Unsplash Random Images Collection
- Used for negative samples (i.e., images that are not chest X-rays).

Workflow

1. Image Verification:

- The API first verifies if the uploaded image is a chest X-ray using the Chest X-ray Detection Model.

2. Pneumonia Prediction:

- If the image is confirmed to be a chest X-ray, it is sent to the Pneumonia Classification Model.
- The model predicts whether pneumonia is present and its type (bacterial or viral).

3. Result Display:

- The prediction results are sent back to the frontend for display.
- Users can then interpret the prediction and act accordingly.

Future Directions

Enhanced Model Accuracy:

Continuous training and fine-tuning of models to improve accuracy and robustness.

Expansion of Dataset:

Incorporating more diverse and extensive datasets to capture a wider range of cases and improve generalization.

About the Author

This project was developed by Tyler Haisman, a computer science student interested in medicine and aspiring to work in technology. For inquiries or collaborations, please contact Tyler via tylerhaisman.com.